

Post-P&R Gate-Level Simulation (GLS) Sign-off

Final_System SoC Design | Post-PnR Timing Verification & Back-Annotated Waveforms

1. Technical Verification Scope & Post-P&R Methodology

Post-Place and Route (Post-PnR) Gate-Level Simulation represents the decisive pre-tapeout physical sign-off milestone. This simulation executes on the fully placed and routed physical netlist back-annotated with actual Standard Delay Format (SDF) timing extracted from real interconnect parasitics (SPEF/RC) and post-Clock Tree Synthesis (CTS) clock distributions. The waveform captures documented herein isolate the primary physical trigger for each operational test scenario, verifying the silicon-accurate behavior of the core under post-route sequential Clock-to-Q latencies, input setup/hold margins, physical interconnect RC propagation delays, and hardware error detection mechanisms without timing violations.

2. Post-P&R Timing Analysis by Test Scenario

[CASE 1A] Register File Synchronous Write Operation

• **Target Subsystem:** Register File Array (regfile) & SYS_CTRL | **Physical Timing Mechanism:** Post-Route Data/Enable Setup Timing & Physical Array Write Latching

Technical Timing Description (Physical Sign-off):

- Captures the post-route synchronous write operation where the active reference clock edge samples the write strobe (WrEn), target address, and input data bus (WrData).
- Demonstrates robust setup and hold margins across back-annotated interconnect paths, verifying glitch-free physical latching into the targeted register flip-flops.

Waveform Screenshot Attachment:

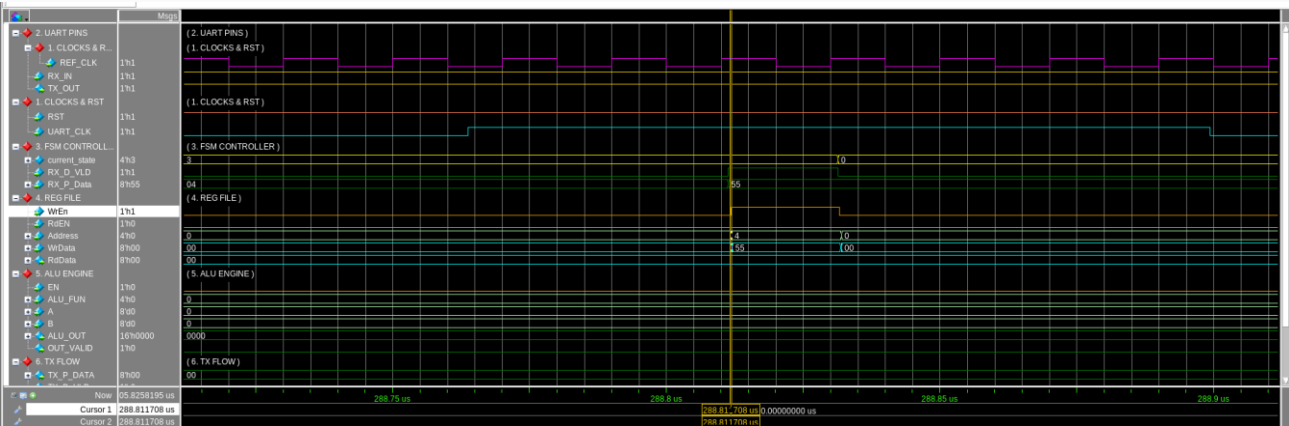


Figure 1A: Case 1A Post-P&R Waveform — Synchronous Memory Write Access and Physical Data Latching.

[CASE 1B] Register File Synchronous Read Access

• **Target Subsystem:** Register File Array (regfile) & SYS_CTRL | **Physical Timing Mechanism:** Post-Route Clock-to-Q (Tcq) with Physical Net Capacitive Loading

Technical Timing Description (Physical Sign-off):

1. Captures the synchronous memory read access as the post-route clock edge triggers the internal register array to drive stored data onto the output bus (RdData).
2. Demonstrates the back-annotated Clock-to-Q propagation delay incorporating actual physical wire routing resistance and capacitive loading on the memory data bus.

Waveform Screenshot Attachment:

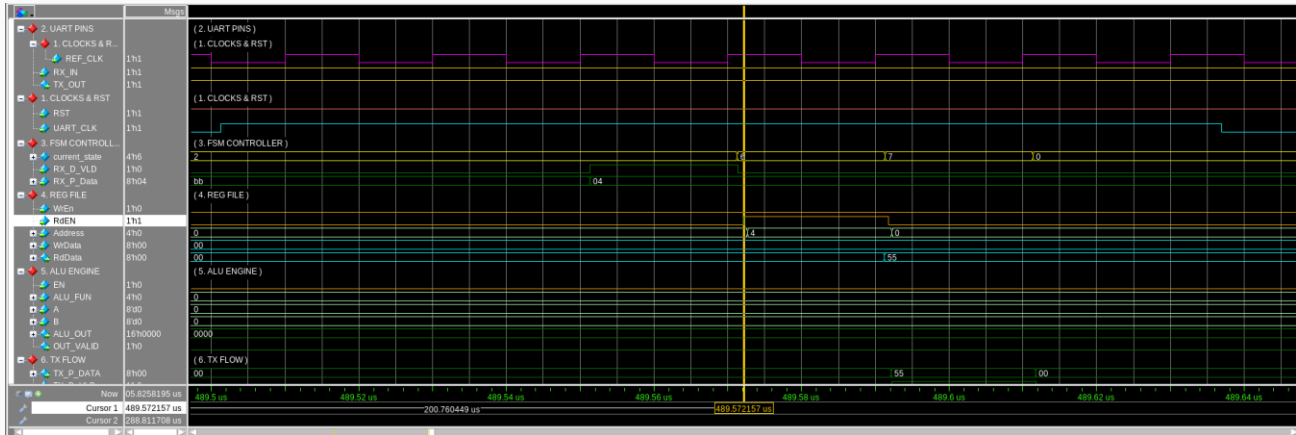


Figure 1B: Case 1B Post-P&R Waveform — Synchronous Memory Read Access and Back-Annotated Clock-to-Q Delay.

[CASE 2] ALU Arithmetic Addition Execution ($10 + 5 = 15$)

• **Target Subsystem:** Arithmetic Logic Unit (ALU Core Adder) | **Physical Timing Mechanism:** Post-Layout Adder Carry-Tree & Interconnect RC Delay

Technical Timing Description (Physical Sign-off):

1. Illustrates the post-route addition operation as operands propagate through the synthesized carry-tree adder network following calculation trigger.
2. Highlights the combined gate and physical interconnect RC delay required for calculation stabilization on ALU_OUT before the valid strobe (OUT_VALID) asserts.

Waveform Screenshot Attachment:

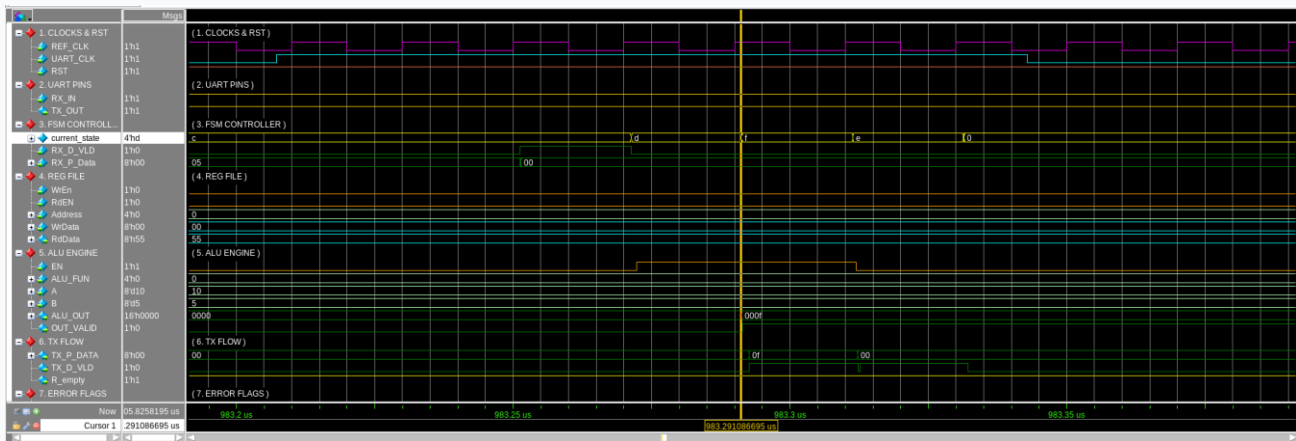


Figure 2: Case 2 Post-P&R Waveform — Adder Interconnect RC Propagation and Output Strobe Assertion.

[CASE 3] ALU Arithmetic Subtraction Execution ($10 - 5 = 5$)

- **Target Subsystem:** Arithmetic Logic Unit (ALU Subtraction Unit) | **Physical Timing Mechanism:** Post-Route Subtractor Datapath & Borrow-Chain Routing Delay

Technical Timing Description (Physical Sign-off):

1. Captures the subtraction datapath execution showing differential logic evaluation and physical borrow-chain propagation across routed ALU macro boundaries.
2. Demonstrates the cumulative standard-cell and wire parasitic delays enabling arithmetic settlement on ALU_OUT and concurrent activation of OUT_VALID.

Waveform Screenshot Attachment:

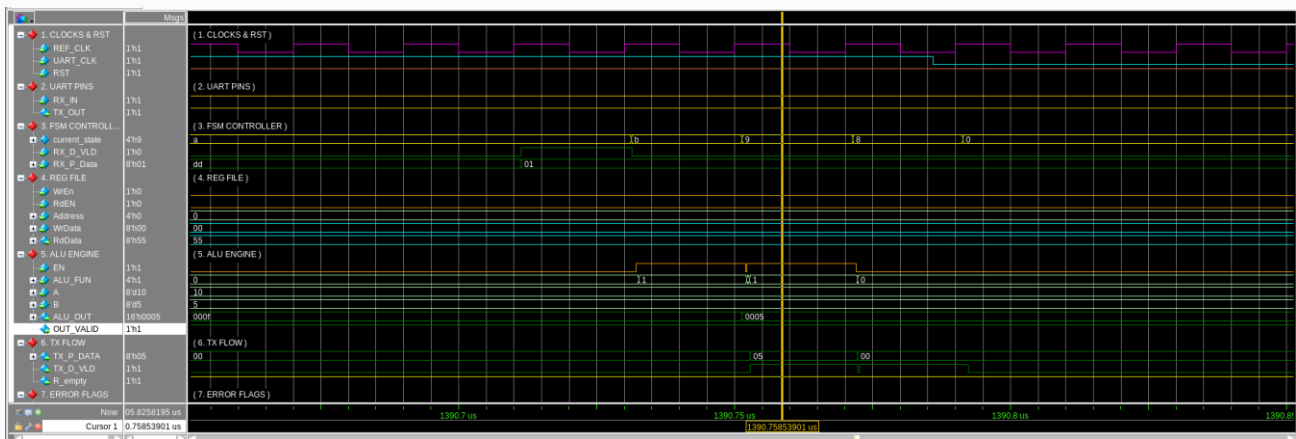


Figure 3: Case 3 Post-P&R Waveform — Subtraction Datapath Evaluation and Borrow-Chain Wire Delay.

[CASE 4] ALU Arithmetic Multiplication Execution ($10 \times 5 = 50$)

- **Target Subsystem:** Arithmetic Logic Unit (Hardware Multiplier Block) | **Physical Timing Mechanism:** Post-P&R Multiplier Critical Path Interconnect & Logic Delay

Technical Timing Description (Physical Sign-off):

- 1. Depicts execution through the physical hardware multiplier array representing the routed design's primary critical path across cascaded partial-product trees.
- 2. Illustrates the extended multi-level routing and cell propagation delay safely resolving to the stable final product on ALU_OUT within allocated clock boundaries.

Waveform Screenshot Attachment:

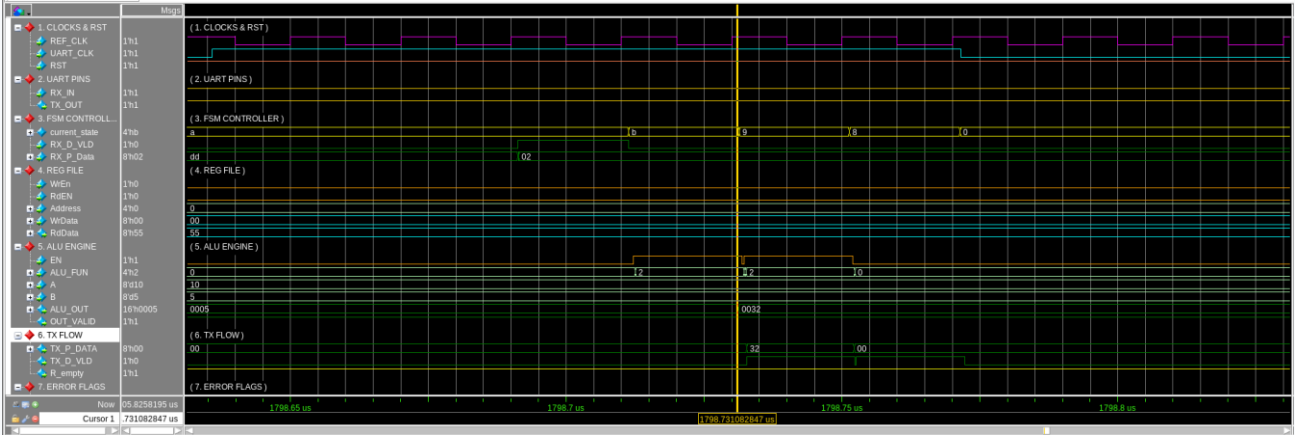


Figure 4: Case 4 Post-P&R Waveform — Multiplier Physical Critical Path and Interconnect Delay.

[CASE 5] UART Receiver Parity Error Detection & Flag Assertion

- **Target Subsystem:** UART Receiver (UART_RX) & Error Detection Logic | **Physical Timing Mechanism:** Post-CTS Clock Domain Parity Sampling & Hardware Flag Latency

Technical Timing Description (Physical Sign-off):

- 1. Captures serial stream parity integrity verification at the post-CTS receiver sampling point following deliberate transmission of a corrupted parity frame.
- 2. Highlights the routed receiver finite state machine transition and physical logic latency leading to the active assertion of the hardware parity error flag (parity_error).

Waveform Screenshot Attachment:

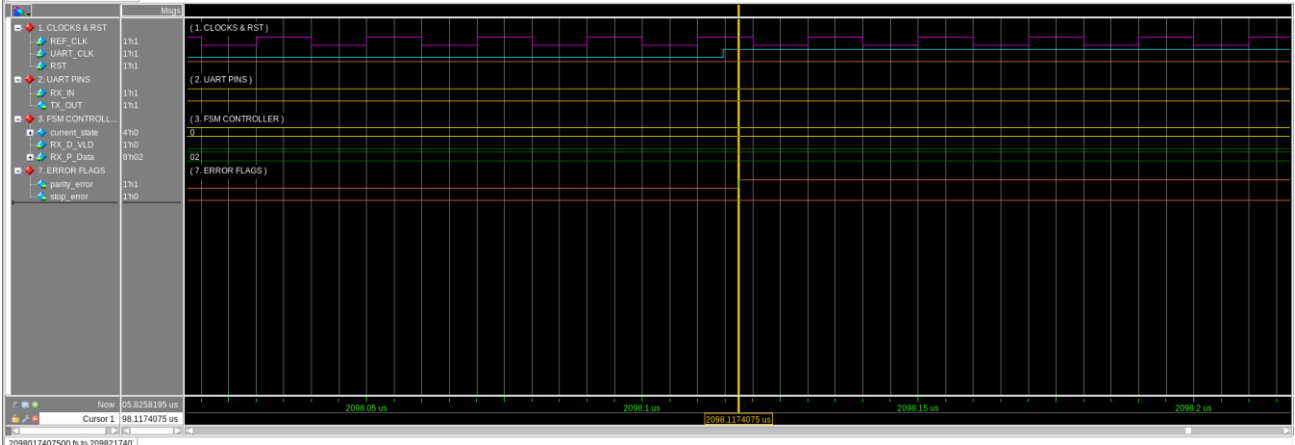


Figure 5: Case 5 Post-P&R Waveform — Post-Route Serial Parity Verification and Flag Latency.

[CASE 6] UART Receiver Framing / Stop Bit Error Detection

• **Target Subsystem:** UART Receiver (UART_RX) & Error Detection Logic | **Physical Timing Mechanism:** Protocol Boundary Check & Framing Detector Routing Latency

Technical Timing Description (Physical Sign-off):

1. Illustrates the post-route receiver frame boundary check evaluating bit timing and detecting an illegal low logic level during the dedicated stop-bit interval.
2. Demonstrates the physical gate propagation and interconnect delay across protocol checking circuitry culminating in the assertion of the framing error flag (stop_error).

Waveform Screenshot Attachment:

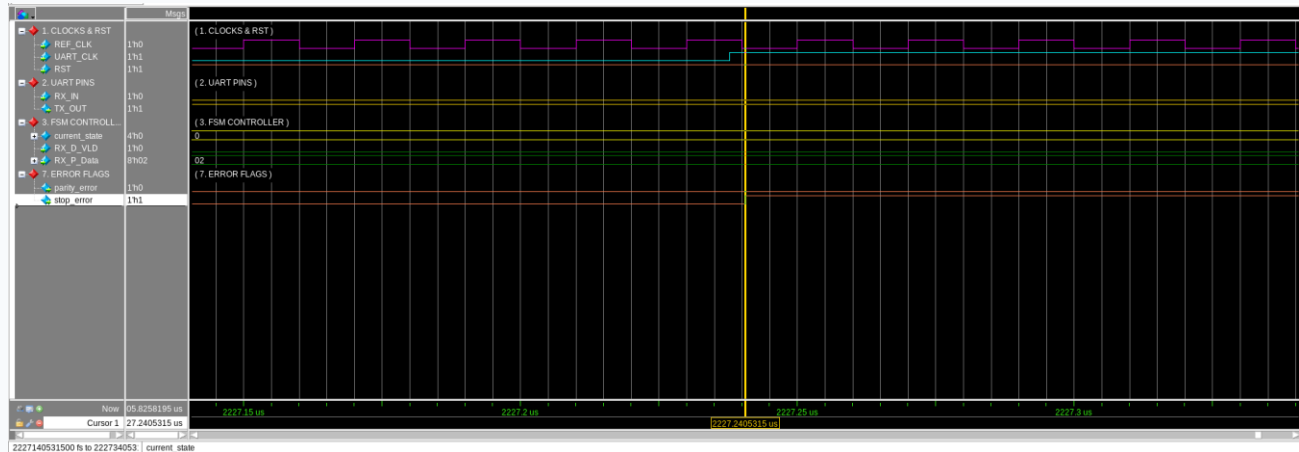


Figure 6: Case 6 Post-P&R Waveform — Frame Boundary Check and Stop Bit Error Detection Latency.